

Autopsy



Autopsy

- **open source digital forensics platform**
 - multiplatform
 - multi-user
- **an extensible framework**
 - collect many open-source tools
 - many underlying operations executed by TheSleuthKit
 - tens for different purposes (acquisition, analysis, reporting, ...)
 - connect them through a Java-GUI

Autopsy workflow

- **create case**

- name of the case
- author(s) meta-data

- **select data sources**

- files, HD images, VM images, ...

- **import data**

- keyword serches, hash analysis, ...

- **data analysis (the big one!)**

- content viewers to display text, images, analytics, ...

- **report generation**

- HTML, XML, ...
 - good if investigators provides extensive remarks/justification during examination

Autopsy import

- **many file systems analysis supported**
 - NTFS, FAT, Ext*, ISO9660, ...
- **interactive provision of the results**
 - as soon as they are found
 - keyword search partial result every few minutes
 - e.g. every 5 minutes
 - prioritization of folders
 - e.g. user folders first

Autopsy default modules

- **recent activity extractor**

- in terms of user action (e.g. web browsing)

- **hash calculation and lookup**

- the hash database ingest service can be configured to use the NIST NSRL hash database of known files

- **file type identification**

- **extension mismatch detector**

- compares a file's actual internal file signature (the "magic bytes") against its outward-facing filename extension

- **embedded file extractor**

- open compressed archive formats and document files to extract their internal contents for analysis

Autopsy default modules

- **exif parser**
- **keyword search**
 - searches all indexed text across the entire data source using strict string matches or regular expressions (Regex)
- **email parser**
 - reads standard email database files (e.g, Outlook *.pst* or *MBOX* formats), parses the messages, and indexes them
- **encryption detection**
 - scans files to calculate their entropy
- **interesting file identifier**
 - files that match specified criteria (e.g. names, path, size, ...)
 - each set has a list of rules, which will match on their chosen file characteristics
 - a file needs to match one rule only to be found

Autopsy default modules

- **correlation engine**

- centralized database module that compares properties (email addresses, hardware IDs, or file hashes) from the current case against historical cases

- **PhotoRec carver**

- **virtual machine extractor**

- detects, opens, and parses virtual machine disk images (such as .vmdk or .vhd) found within the main data source

- **data source integrity**

- Computes & verifies the cryptographic hashes of the entire physical/logical data source at the beginning and end of ingest

- **android analyzer**

- parsing script that ingests Android database structures to acquire SMS messages, call logs, contacts, application data

NIST NSRL

- **National Software Reference Library**
 - supported by NIST...
 - ... and U.S. Department of Homeland Security, FBI, state and local law enforcement
- ***“promote efficient and effective use of computer technology in the investigation of crimes involving computers”***
- **collect software profiles from various sources**
- **build a Reference Data Set (RDS) of information**
 - i.e. a collection of digital signatures of known software applications

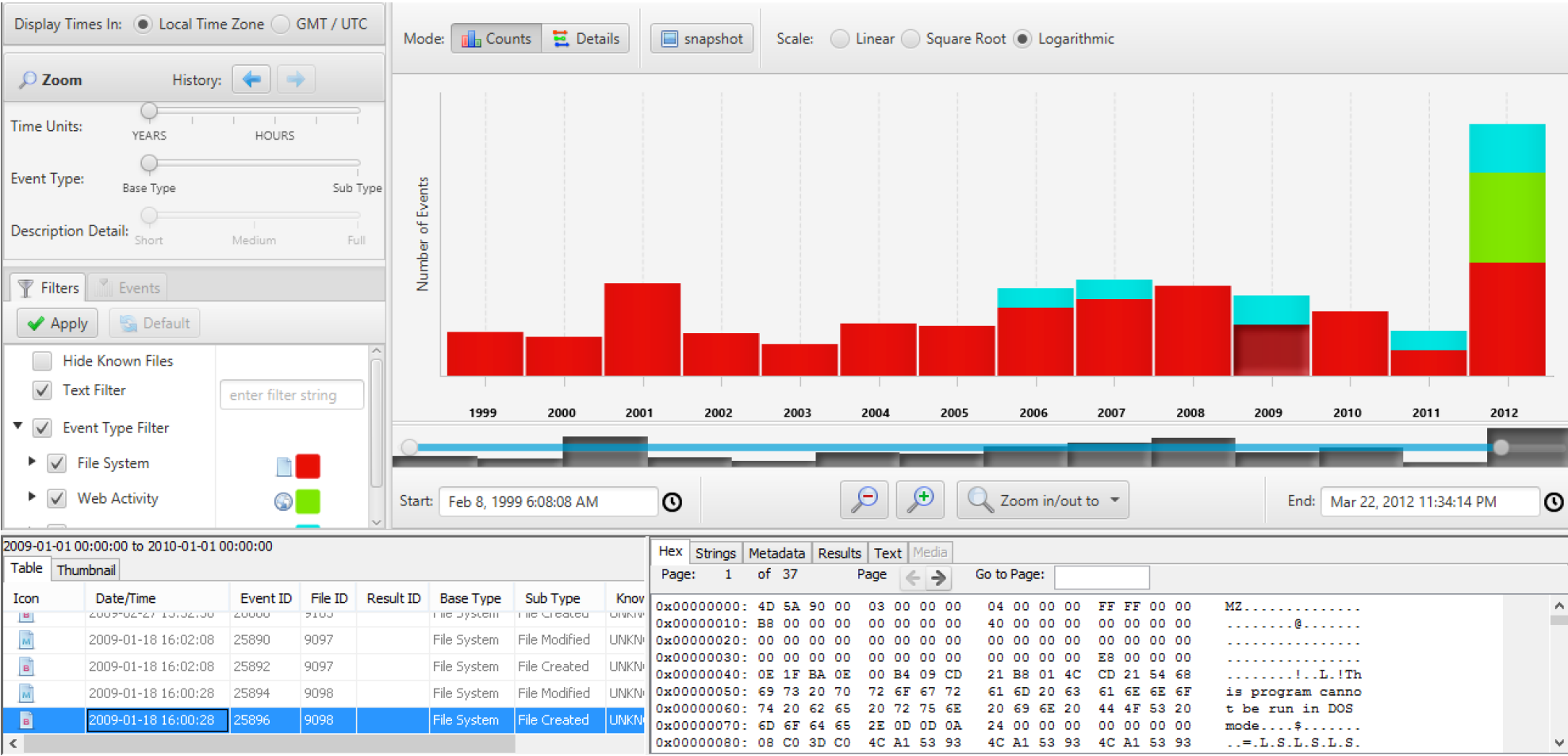
NSRL categorization

- **targets free and popular apps**
- **tagged with**
 - application title
 - version
 - vendor
 - OS
 - file path
 - extension,
 - size
- **categorized by “type”**
 - e.g. word proccession, encryption
 - not by “maliciousness” (no “known good”)

Timeline

- **a central visual tool**
 - when a computer is used?
 - what events occurred before/after another one?
- **gather timestamps from**
 - files
 - web artifacts
 - meta-data (EXIF, GPS, ...)

Timeline



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